Junyao Dong

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Education

Nanyang Technological University — Singapore
M.Eng. by Research in Computing and Data Science Engineering — 08/2024 – 06/2026
- GPA: 4.17/5.0. Co-advised by Cheng Long and Shuhao Zhang.
- **Research Interests:** Vector databases, retrieval systems, and machine learning systems.

The University of Manchester — Manchester, UK
Visiting Student in Computer Science — 08/2019 – 05/2020
- Overall Average: 70% (First-Class Standard).

Guangdong University of Foreign Studies — Guangzhou, China
B.Eng. in Computer Science — 09/2017 – 06/2021
- GPA: 3.4/4.0.

Publications (* = Equal contribution)

The Program Counter Is a Path: Code, Data, and Control in One Addressable Tree
Peng Li*, Junyao Dong*
- Submitted to *Proceedings of the ACM on Programming Languages*, 2027 (**PLDI'27**). [Website] [Code]

Shared Memory in Multi-Agent Systems: Which View Does the Agent Read?
Junyao Dong, Xisen Wang, Yu Chen, Yi Li, Qi Zhang
- Submitted to the *26th International Conference on Autonomous Agents and Multiagent Systems* (**AAMAS'27**).

ANNchor: Concurrency Control for Freshness in Streaming Graph-Based Approximate Nearest Neighbor Search
Junyao Dong, Shuhao Zhang, Wenchao Zhou, Cheng Long
- Revision at *Proceedings of the ACM on Management of Data*, 2027 (**SIGMOD'27**, Round 3; decision Dec 12)[1]. [Latest Status]

CANDOR-Bench: Benchmarking In-Memory Continuous ANNS under Dynamic Open-World Streams
Mingqi Wang*, Junyao Dong*, Zhuoyan Wu, Jun Liu, Ruicheng Zhang, Jianjun Zhao, Ruipeng Wan, Xinyan Lei, Shuhao Zhang, Bolong Zheng, Haikun Liu, Xiaofei Liao, Hai Jin
- *Proceedings of the ACM on Management of Data*, 2026 (**SIGMOD'26**). [Paper] [Code]

Research Experience († = Full-time)

Vector Database Group @ NTU — Singapore
Research Associate† (Advised by Cheng Long) — 08/2026 – Present
- Applying vector search and RaBitQ quantization to long-context LLM inference, studying KV-cache retrieval for sparse attention in decoding.

PageIndex — Remote
Research Intern (Co-advised by Mingtian Zhang and Yu Tang) — 12/2025 – Present
- Led the design and implementation of ConDB, a chunk-free, vector-free context database for retrieval over long, hierarchically structured documents.
- Collaborated with MongoDB on storing and searching large hierarchical JSON in the MongoDB Python driver.

Vector Database Group @ NTU & IntelliStream @ HUST — Singapore
Graduate Student Researcher† (Co-advised by Cheng Long and Shuhao Zhang) — 08/2024 – 06/2026
- Designed ANNchor to keep streaming vector search fresh: post-acknowledgment visibility on one live HNSW graph, with up to 4.6× search throughput over global locking and 68% lower P99 latency under write-heavy load.
- Co-developed CANDOR-Bench, a continuous ANNS benchmark of 19 algorithms on 12 datasets under drift and concurrent read–write streams; co-designed the streaming evaluation harness.

[1] SIGMOD reviews papers in rolling rounds. For Round 3, the first notification (accept, reject, or revise) was on Oct 19, 2026; the revised paper is due Nov 19, and the final accept/reject decision is announced on Dec 12, 2026. The latest status will be posted on my homepage.

The Hong Kong University of Science and Technology (Guangzhou) Guangzhou, China
Research Assistant† (Advised by Ying Cui) 04/2024 – 06/2024

- Developed an adaptive bitrate system for video delivery under variable network conditions that cuts playback buffering without losing visual quality.

Selected Open-Source Projects

RaBitQ-Library ∼300 GitHub stars
Contributor github.com/VectorDB-NTU/RaBitQ-Library

- Hardened IVF index construction in the official library for RaBitQ, a randomized bitwise quantization method for nearest neighbor search used by Meta, NVIDIA, Microsoft, and Alibaba; prototyped quantized graph-index search integration.

RL-Kernel 400+ GitHub stars
Contributor github.com/RL-Align/RL-Kernel

- Implemented CUDA/Triton kernels for DeepSeek V4 compressed attention (one-softmax MQA with an attention sink, grouped output projection) in an RL post-training stack for operator-level train–inference consistency.

vllm-rlt 200+ GitHub stars
Contributor github.com/InferMatrixLab/vllm-rlt

- Designed the engine loop of a looped-language-model inference engine that batches requests at different loop depths together; synchronous and asynchronous steps share scheduling and result handling, while submission, GPU completion, and output delivery stay separate.

Industry Experience († = Full-time)

Baidu Shanghai, China
Software Engineer†, Cloud Services 05/2021 – 03/2024

- Built ingest and storage for Baidu Cloud Live Streaming Service, integrating H.264/H.265 at the edge and writing outputs as MP4, FLV, and HLS.
- Built on-air production for Baidu Cloud Live Cast Production Studio, adding watermarks, picture-in-picture, transitions, and subtitles on a customized FFmpeg engine and Baidu's RTC stack.

Momenta Suzhou, China
Research and Development Intern, Autonomous Driving 10/2020 – 05/2021

- Developed log infrastructure for replaying recorded drives as training data, synchronizing perception and planning streams onto one timeline.

Awards and Honors

Second Prize, National Modeling Competition 2020

Overseas Student Programme Award 2020

Academic Service

Availability and Reproducibility Committee (ARC)

- SIGMOD 2026 2026

External Reviewer

- VLDB 2026, SIGMOD 2027, VLDB 2027, WSDM 2027 2026
- ICDE 2026 2025